

Section A

Answer **all** questions in the spaces provided.

- 1 Describe the single transformation which maps the curve with the equation

$$y = \ln x$$

onto the curve with the equation

$$y = 2 \ln x$$

Tick (✓) **one** box.

[1 mark]

Stretch, scale factor 2, parallel to the y-axis

☒

Stretch, scale factor 2, parallel to the x-axis

☐

AQA PAPER 2
2025 A LEVEL
MARK
SCHEME

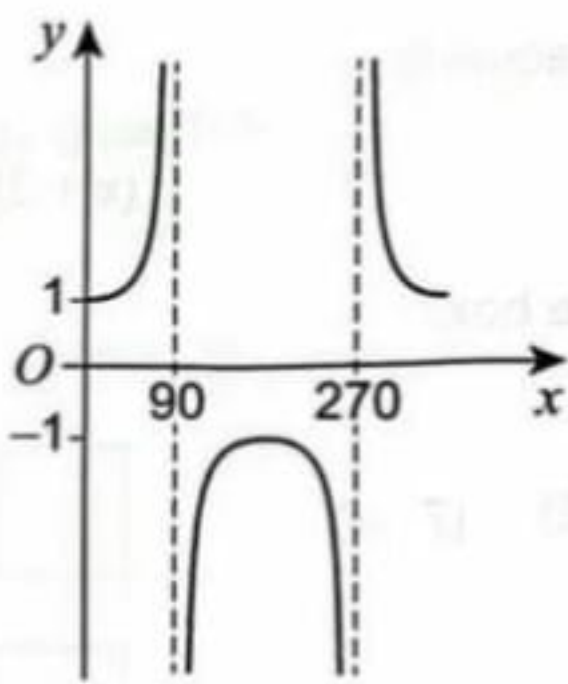
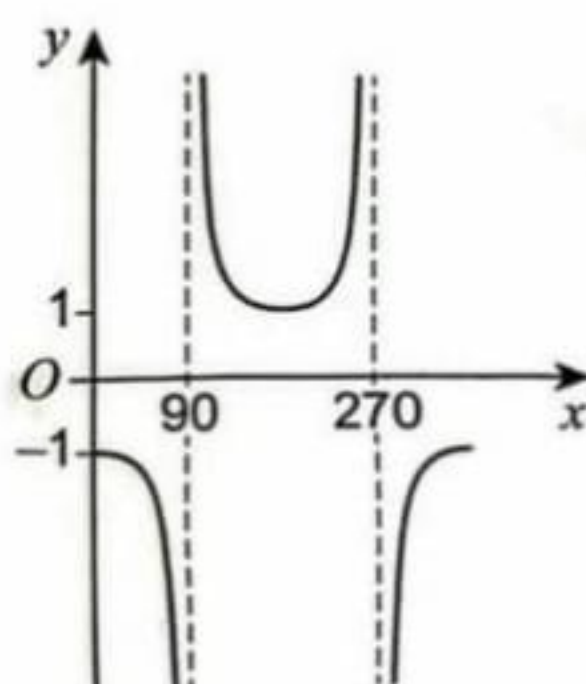
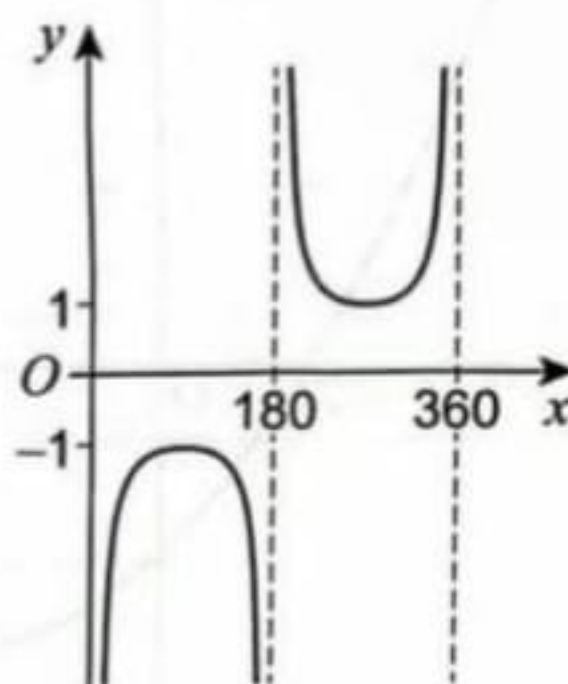
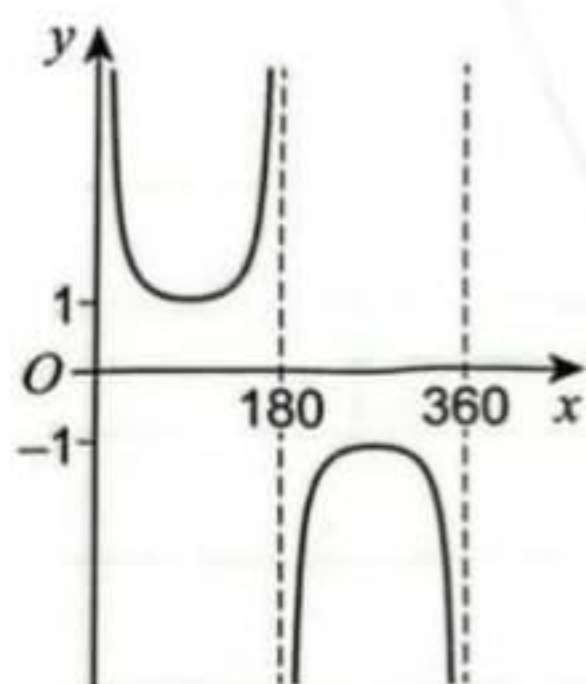
2

One of the diagrams below shows the graph of $y = \operatorname{cosec} x^\circ$ for $0 \leq x \leq 360$

Identify the correct graph.

Tick (✓) **one** box.

[1 mark]



Turn over for the next question

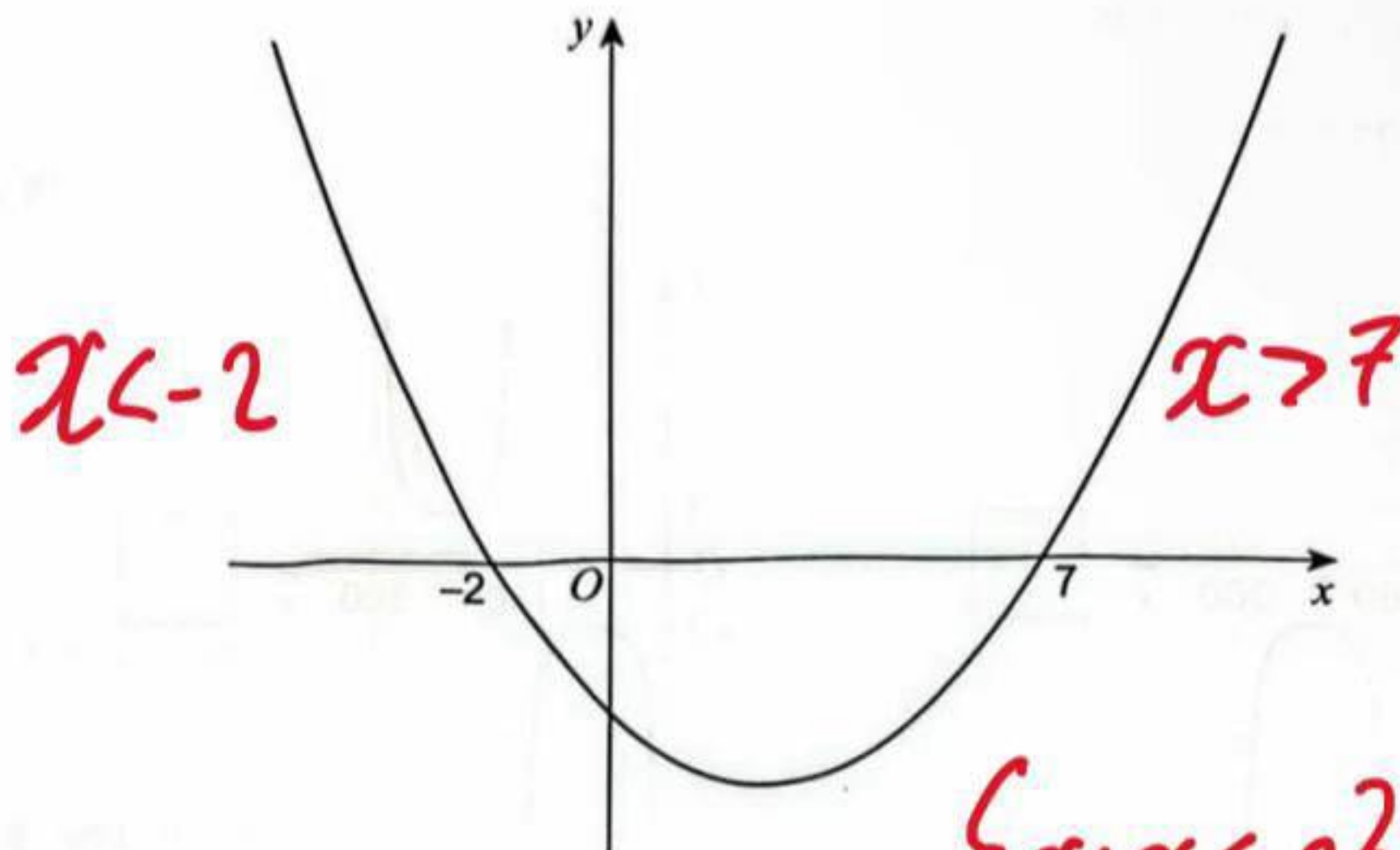
Vertical asymptotes where $\sin x = 0$

$x = 0, 180$

between 0 and 90 $\sin$ is positive

3

The diagram shows the graph with equation $y = (x + 2)(x - 7)$



Solve the inequality.

$$(x + 2)(x - 7) > 0$$

Tick (✓) **one** box.

[1 mark]

$x \in (-\infty, -2) \cap (7, \infty)$

☐

$x \in (-\infty, -2) \cup (7, \infty)$

☒

$x \in (-\infty, -2] \cap [7, \infty)$

☐

$x \in (-\infty, -2] \cup [7, \infty)$

☐

Solve the equation

$$5^{x-1} = 20$$

Give your answer in an exact form.

[2 marks]

$$\log_5(5^{x-1}) = \log_5 20$$

$$x-1 = \log_5 20$$

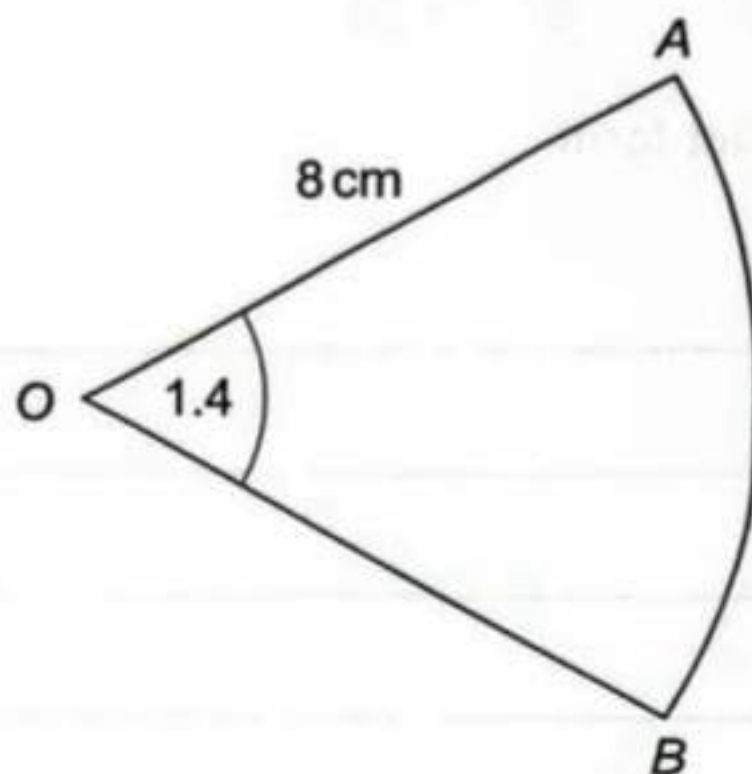
$$x = \log_5 20 + 1$$

$$\text{or } \log_5(100)$$

Turn over for the next question

5

The diagram shows a sector AOB of a circle with centre O and radius 8 cm.



$$\frac{1}{2}r^2\theta$$

The angle AOB is 1.4 radians.

Find the area of the sector.

[2 marks]

$$\frac{1}{2} \times 8^2 \times 1.4$$

$$= 44.8$$

- 6 (a) (ii) Use the trapezium rule with six ordinates (5 strips) to find an approximate value for the area of the shaded region.

Give your answer to five significant figures.

[3 marks]

$$\frac{h}{2} [y_0 + y_n + 2(y_1 + y_2 + \dots)]$$

$$\frac{0.8}{2} [7.3891 + 10.9196 + 2(6.124 + 6.3249 + \dots)]$$

$$= 30.026 \text{ (5sf)}$$

- 6 (b) A student finds an improved approximation for the area of the shaded region by using the trapezium rule with 11 ordinates.

The student correctly obtains 29.759 as their improved approximation.

The student claims that the exact area must be greater than 29.759

Without further calculation, explain whether or not the student is correct.

[2 marks]

the Curve is Convex, any pair of lines will be connecting above the Curve giving an over-estimate. He is wrong.

7 The point A lies on the curve with equation

$$y = x^3 + px^2 + qx + 12$$

7 (a) Given that A has coordinates $(-5, 37)$, show that

$$5p - q = 30$$

$$37 = (-5)^3 + p(-5)^2 + q(-5) + 12$$

[2 marks]

$$37 = -(125 + 25p - 5q + 12)$$

$$25p - 5q = 150$$

$$5p - q = 30$$

7 (b) Given that A is a stationary point, show that

$$10p - q = 75$$

[3 marks]

$$\frac{dy}{dx} = 3x^2 + 2px + q = 0 \text{ at } x = -5$$

$$3(-5)^2 + 2p(-5) + q = 0$$

$$75 - 10p + q = 0$$

$$10p - q = 75$$

7 (c) Hence find the value of p and the value of q

Simultaneous

[1 mark]

$$\begin{aligned} 5p - q &= 30 \\ 10p - q &= 75 \end{aligned}$$

$$5p = 45 \quad p = 9 \quad q = 15$$

7 (d) The curve with equation

$$y = x^3 + px^2 + qx + 12$$

has a second stationary point B

Find the coordinates of B

Fully justify your answer.

[3 marks]

$$x^3 + 9x^2 + 15x + 12$$

$$\frac{dy}{dx} = 3x^2 + 18x + 15 = 0$$

We know $x = -5$ is a stationary point, so $(x+5)$ or $(3x+15)$ factor

$$3x^2 + 18x + 15 = (x+5)(3x+3) = 0$$

$$x = -1 \quad y = 5$$

8 (a)

Given that $\cos x \neq 0$ state the value of $\sec^2 x - \tan^2 x$

[1 mark]

$$\tan^2 x + 1 = \sec^2 x$$

$$(\tan^2 x + 1) - \tan^2 x = 1$$

8 (b)

Show that

$$(3 \sec x + 5 \tan x)(5 \sec x - 3 \tan x) - \frac{16 \tan x}{\cos x} = N$$

where N is an integer to be found.

[4 marks]

expand

$$15 \sec^2 x + 16 \sec x \tan x - 15 \tan^2 x - \frac{16 \tan x}{\cos x}$$

$$15(\sec^2 x - \tan^2 x) + 16 \sec x \tan x - \frac{16 \tan x}{\cos x}$$

1

 $\frac{1}{\cos x}$ $\frac{\sin x}{\cos x}$

$$15 + \frac{16 \sin x}{\cos^2 x} - \frac{16 \sin x}{\cos^2 x}$$

$$= 15$$

8 (c)

State the two values of x between 0° and 360° for which the value of N in part (b) is not valid.

[1 mark]

Invalid when $\cos x = 0$

$$x = 90^\circ \quad x = 270^\circ$$

Turn over for the next question



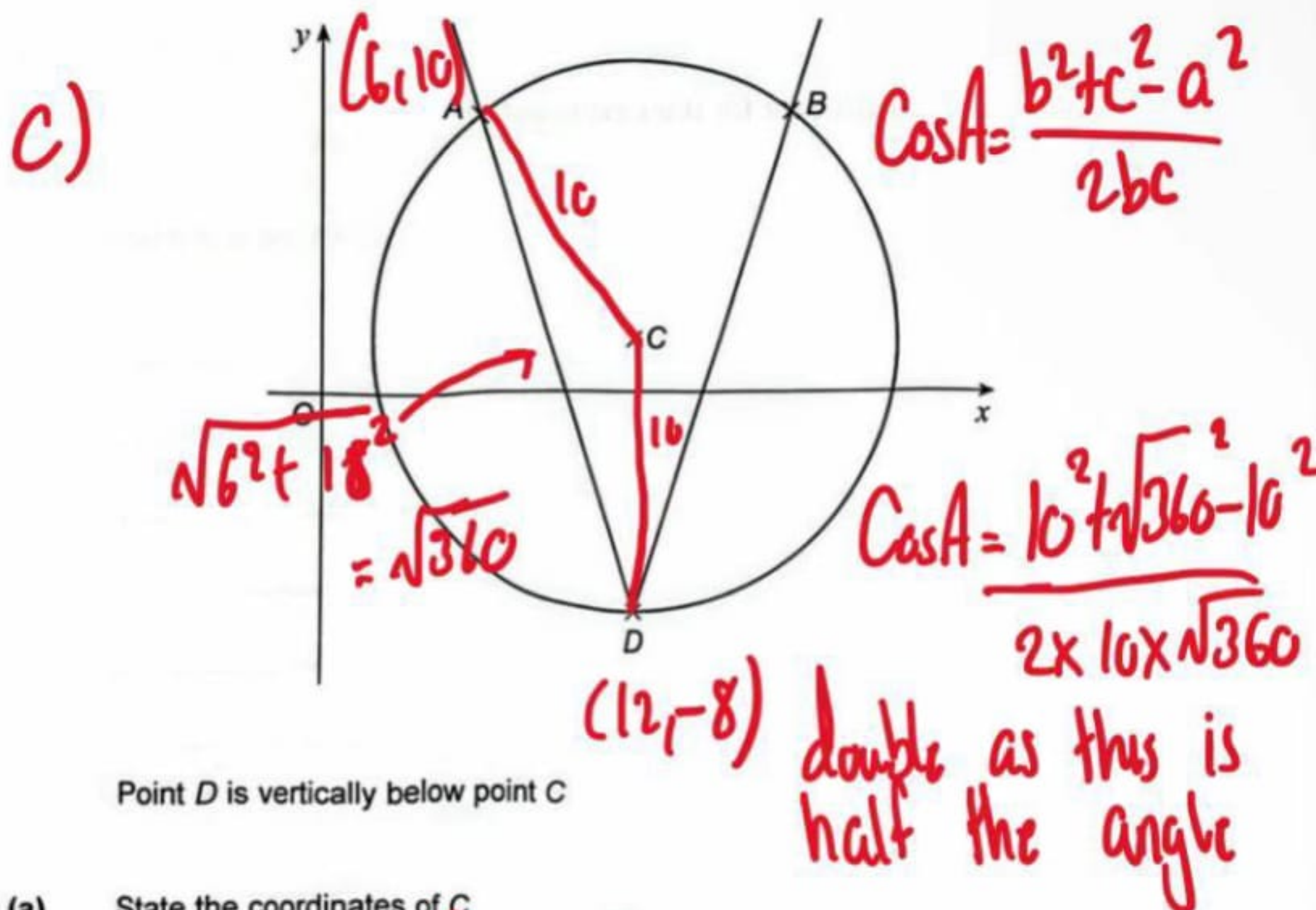
9

A circle with centre C has equation

$$(x - 12)^2 + (y - 2)^2 = 100$$

The graph with equation

$$y = |3x - 36| - 8$$

intersects the circle at the points A , B and D as shown in the diagram.Point D is vertically below point C

9 (a)

State the coordinates of C

$(12, 2)$

[1 mark]

9 (b)

State the coordinates of D

$(12, -8)$

[1 mark]

9 (c) The coordinates of A are $(a, 10)$

Find the value of a

Fully justify your answer.

[3 marks]

on Circle so - Satisfies Circle eqn

$$(x-12)^2 + (y-2)^2 = 100$$

$$(a-12)^2 + 8^2 = 100$$

$$(a-12)^2 = 36$$

$$a = \pm 6 + 12 \quad \text{Since } a - 12 = \pm 6 \quad \text{Since } a \text{ is more left } a = 6$$

9 (d) (i) Find, in radians, the angle ADB

Give your answer to **three** significant figures.

[2 marks]

0.644, See previous page for working

9 (d) (ii) Hence or otherwise find the length of the minor arc AB

Give your answer to **three** significant figures.

[2 marks]

we can use Circle Theorems to arrive at the fact angle ACB = 2X Angle ADB

$$\text{angle ACB} = 1.288 \quad r\theta = 10 \times 1.288 = 12.9 (3sf)$$

$$y = x^k \ln x \text{ for } x > 0$$

where k is a positive integer.

$$v \frac{du}{dx} + u \frac{dv}{dx}$$

10 (a) Show that

$$\frac{dy}{dx} = x^{k-1} [A + k \ln x]$$

where A is a constant to be found.

[4 marks]

$$x^k \left(\frac{1}{x} \right) + \ln x (k x^{k-1})$$

$$x^k (x^{-1}) + \ln x (k x^{k-1})$$

$$x^{k-1} + \ln x (k x^{k-1})$$

$$= x^{k-1} (1 + k \ln x)$$

10 (b) Hence show that the y -coordinate of the stationary point of C can be written as $-\frac{1}{ke}$

Fully justify your answer.

[5 marks]

$$\frac{dy}{dx} = 0 \text{ for } x \text{ coord}$$

$$x^{k-1} (1 + k \ln x) = 0 \quad 1 + k \ln x = 0 \quad x = e^{-1/k}$$

$$y = (e^{-1/k})^k \ln(e^{-1/k}) = e^{-1} \times -\frac{1}{k} = -\frac{1}{ke}$$

- 10 (c) Given that the stationary point of C has coordinates $\left(\frac{1}{e}, -\frac{1}{e}\right)$ state the value of k

$$\frac{-1}{ke} = -\frac{1}{e} \quad k=1$$

[1 mark]

- 10 (d) Prove that C does **not** have a point of inflection.

[3 marks]

with $k=1$, $\frac{dy}{dx} = 1 + \ln x$

$$\frac{d^2y}{dx^2} = \frac{1}{x} \quad \text{for } x > 0 \quad \text{the}$$

Sign of second derivative never changes (stays +), therefore no point of inflection.

11

An object is pulled along a horizontal surface. The normal reaction between the surface and the object is 125 newtons.

The object experiences a constant frictional force of 40 newtons.

Find the value of the coefficient of friction.

Circle your answer.

[1 mark]

0.245

0.32

0.5

3.125

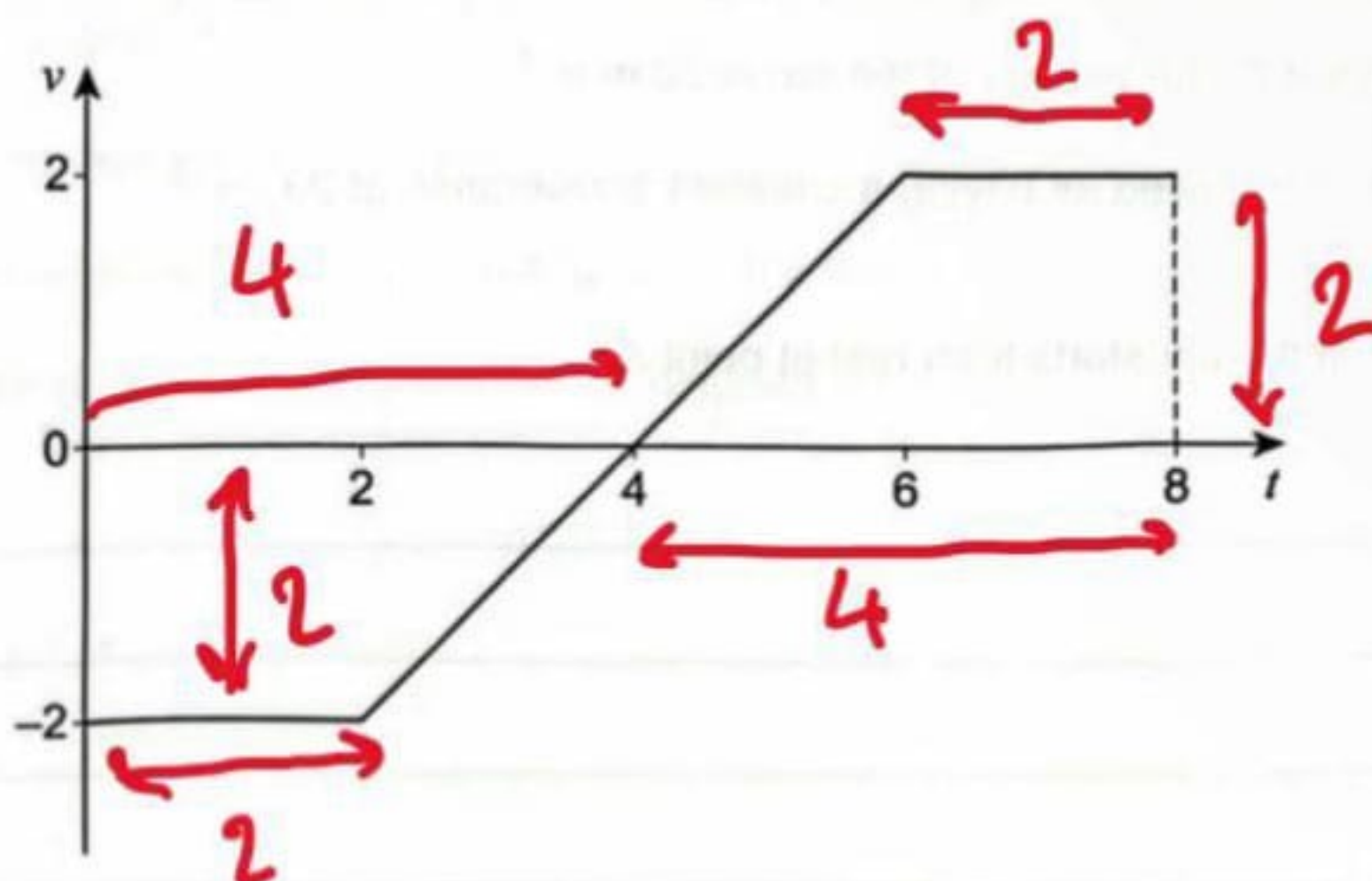
$$F = \mu R$$

$$40 = \mu \times 125$$

$$\mu = \frac{40}{125} =$$

A particle moves in a straight line for 8 seconds.

The velocity v , in m s^{-1} , of the particle at time t seconds is shown in the diagram.



Find the displacement, in metres, of the particle after 8 seconds.

Circle your answer.

[1 mark]

-12

0

6

12

Turn over for the next question

two equal trapeziums,
different directions

13

A car moves, in a straight line on a horizontal road, from a point A to a point B

The distance AB is 225 metres.

At the point B, the velocity of the car is 30 m s^{-1}

The car is modelled as having a constant acceleration of 2 m s^{-2}

13 (a)

Show that the car starts from rest at point A

$$S=225 \quad u=u \quad v=30 \quad a=2 \quad t=t$$

[2 marks]

$$v^2 = u^2 + 2as$$

$$30^2 = u^2 + 2 \times 225 \times 2$$

$$900 = u^2 + 900 \quad u^2 = 0 \quad u = 0$$

13 (b)

The car continues to move past the point B in the same straight line.

Explain why the model would eventually become invalid.

[1 mark]

Acceleration will not increase
forever

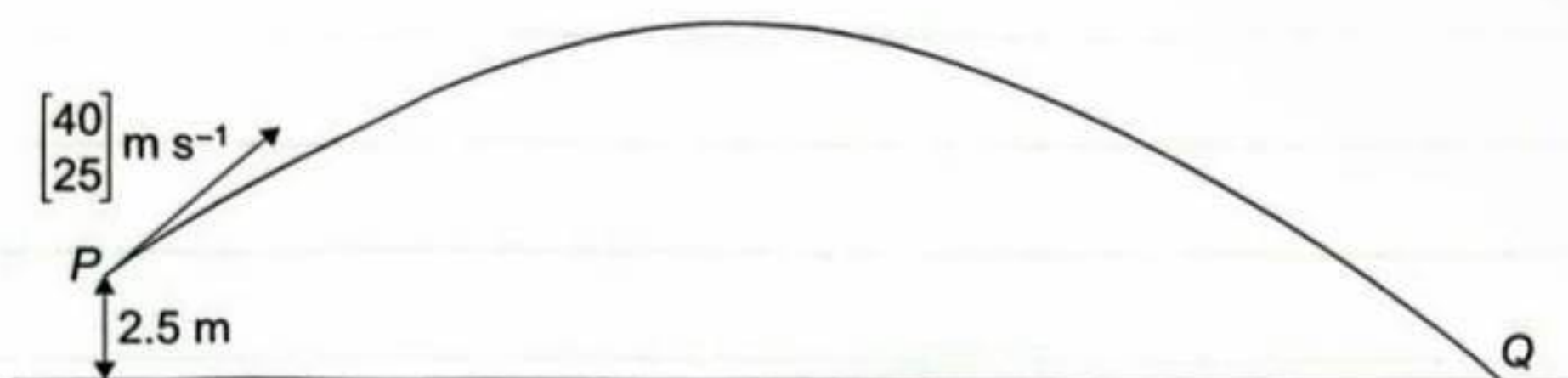
In this question use $g = 9.8 \text{ m s}^{-2}$

An arrow is projected from a point P which is at a height of 2.5 metres above the horizontal ground.

The arrow has an initial velocity of $\begin{bmatrix} 40 \\ 25 \end{bmatrix} \text{ m s}^{-1}$

The arrow lands on the horizontal ground at a point Q

The path of the arrow is shown in the diagram.



Find the speed of the arrow at point Q

[4 marks]

$$S = \begin{pmatrix} sx \\ -2.5 \end{pmatrix} \quad u = \begin{pmatrix} 40 \\ 25 \end{pmatrix} \quad v = \begin{pmatrix} 40 \\ v_y \end{pmatrix} \quad a = \begin{pmatrix} 0 \\ -9.8 \end{pmatrix} \quad t = t$$

$$v^2 = u^2 + 2as$$

$$\begin{pmatrix} 40^2 \\ v_y^2 \end{pmatrix} = \begin{pmatrix} 40^2 \\ 25^2 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ -9.8 \end{pmatrix} \begin{pmatrix} sx \\ -2.5 \end{pmatrix}$$

$$v_y^2 = 625 + 49 = 674$$

$$\text{Speed} = \sqrt{v_x^2 + v_y^2} = \sqrt{674 + 1600} = 47.7$$

15

A particle moves under the actions of two forces, F_1 and F_2

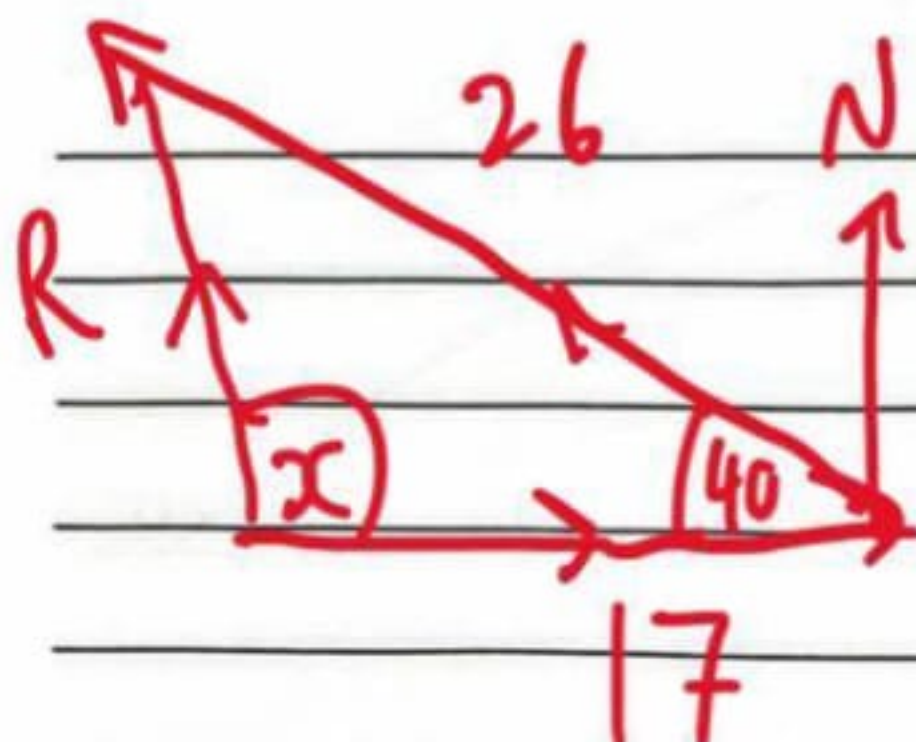
F_1 has magnitude 17 newtons and acts due East.

F_2 has magnitude 26 newtons and acts at a bearing of 310°

The resultant of F_1 and F_2 is R

15 (a) Show that the magnitude of R is 17.0 newtons, correct to **three** significant figures.

[4 marks]



$$R^2 = 26^2 + 17^2 - (2 \times 26 \times 17 \times \cos 40)$$

$$R^2 = 287.8167 \dots$$

$$R = \sqrt{ANS} = 17 \text{ (3SF)}$$

15 (b) Find the angle that R makes with F_1

Give your answer to the nearest degree.

[2 marks]

$$\frac{\sin x}{26} = \frac{\sin 40}{17}$$

$$x = \sin^{-1} \left(\frac{26 \sin 40}{17} \right)$$

$$x = 79^\circ \text{ (nearest 1)}$$

15 (c) A third force, F_3 , acts upon the particle so that the particle is in equilibrium.

15 (c) (i) State the magnitude of F_3

[1 mark]

In order to bring the particle
into equilibrium its magnitude must be
the same. 17.

15 (c) (ii) State the bearing on which F_3 acts.

[1 mark]

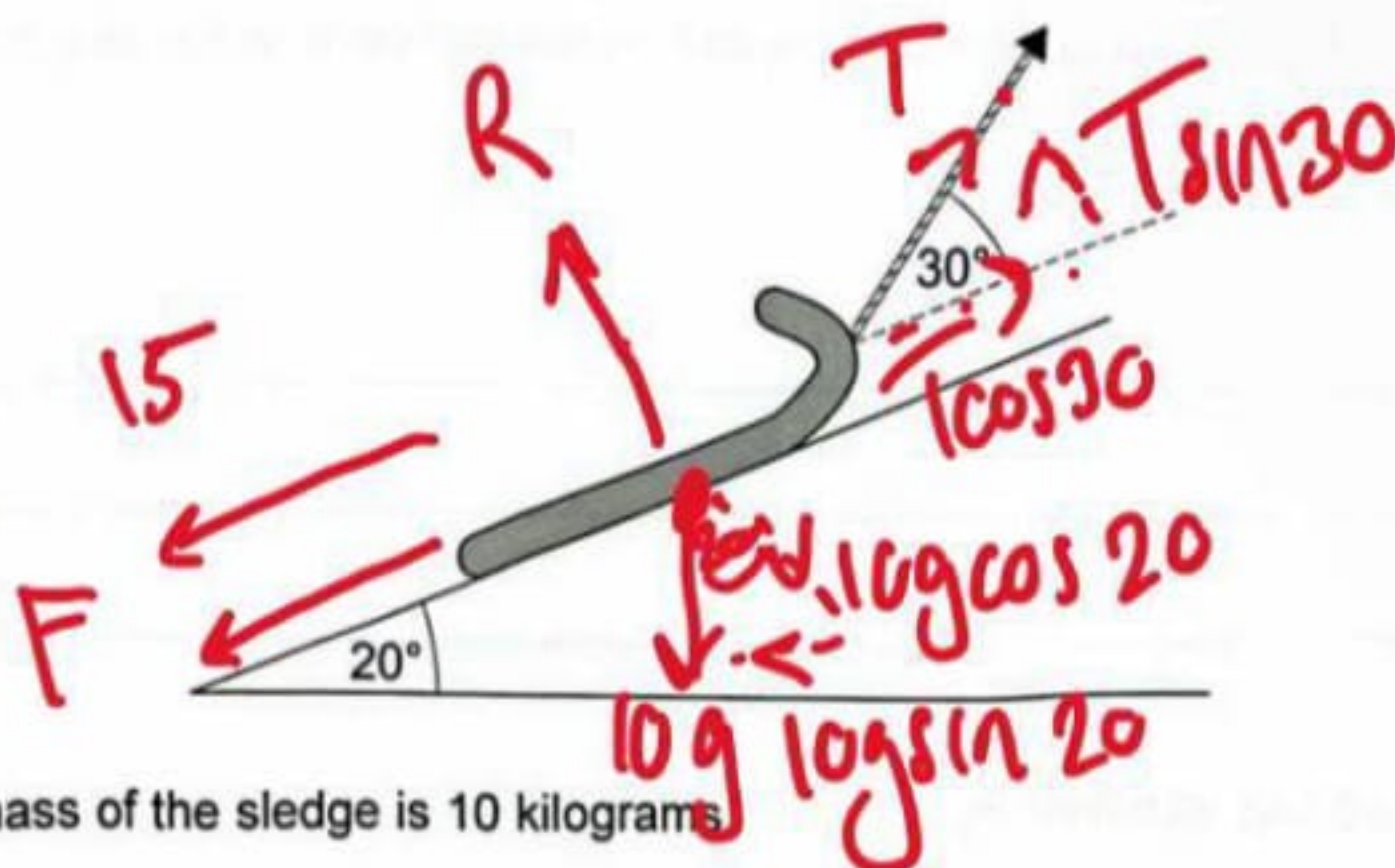
168°

In this question use $g = 9.8 \text{ m s}^{-2}$

A sledge is pulled in a straight line up a rough path, as shown in the diagram.

The path is inclined at an angle of 20° to the horizontal.

The sledge is pulled by a light, inextensible rope inclined at an angle of 30° to the path.



The mass of the sledge is 10 kilograms

- 16 (a) In one model, the sledge moves at a constant speed and experiences a combined resistance force of 15 newtons.

Find the tension in the rope for this model.

$a = 0$

[4 marks]

Handwritten solution:

$$R + T \sin 30$$

$$15 + 10g \sin 20$$

$$10g \cos 20$$

$$T \cos 30$$

$$F = ma$$

$$T \cos 30 - 15 - 10g \sin 20 = 0$$

$$T = \frac{10g \sin 20 + 15}{\cos 30}$$

$$T = 56.0 \text{ (3sf)}$$

16 (b)

In a different model, the sledge experiences no air resistance.

The tension in the rope for this model is 54 N

The coefficient of friction between the sledge and the path is 0.2

Find the acceleration of the sledge for this model.

[7 marks]

$R + 54 \sin 30$
 $54 \cos 30$
 $F + 10g \sin 20$
 $10g \cos 20$
 $F_{\text{net}} = ma$
 $54 \cos 30 - F - 10g \sin 20 = 10a$
 $a = \frac{54 \cos 30 - F - 10g \sin 20}{10}$

need F , $F = \mu R$ $R + 54 \sin 30 = 10g \cos 20$
 $R = 10g \cos 20 - 54 \sin 30$ $\mu R = 0.2(10g \cos 20 - 54 \sin 30)$
 $a = \frac{54 \cos 30 - 0.2(10g \cos 20 - 54 \sin 30) - 10g \sin 20}{10}$
 $a = 0.023 \text{ (3sf)}$

16 (c)

State an assumption you have used to answer both parts (a) and (b).

[1 mark]

Sledge is a particle

A triangle has vertices A, B and C with position vectors given by

$$\vec{OA} = \begin{bmatrix} -3 \\ 5 \\ 1 \end{bmatrix}, \vec{OB} = \begin{bmatrix} 5 \\ 4 \\ -2 \end{bmatrix} \text{ and } \vec{OC} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

17 (a) (i) Find $\vec{AB}$

[1 mark]

$$\vec{AB} = \vec{AO} + \vec{OB}$$

$$= \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix} + \begin{pmatrix} 5 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 8 \\ -1 \\ -3 \end{pmatrix}$$

17 (a) (ii) Find the magnitude of $\vec{AB}$

[1 mark]

$$\sqrt{8^2 + 1^2 + 3^2} = \sqrt{74}$$

17 (b) Hence show that triangle ABC is a scalene triangle.

[5 marks]

$$\vec{BC} = \vec{BO} + \vec{OC} = \begin{pmatrix} -4 \\ -5 \\ 2 \end{pmatrix}$$

$$\vec{AC} = \vec{AO} + \vec{OC} = \begin{pmatrix} 4 \\ -6 \\ -1 \end{pmatrix}$$

3 diff lengths

$$|BC| = \sqrt{4^2 + 5^2 + 2^2} = \sqrt{45}$$

$$|AC| = \sqrt{4^2 + 6^2 + 1^2} = \sqrt{53}$$

In this question use $g = 9.81 \text{ m s}^{-2}$

A uniform platform AD has length 2.5 metres and weight 400 newtons.

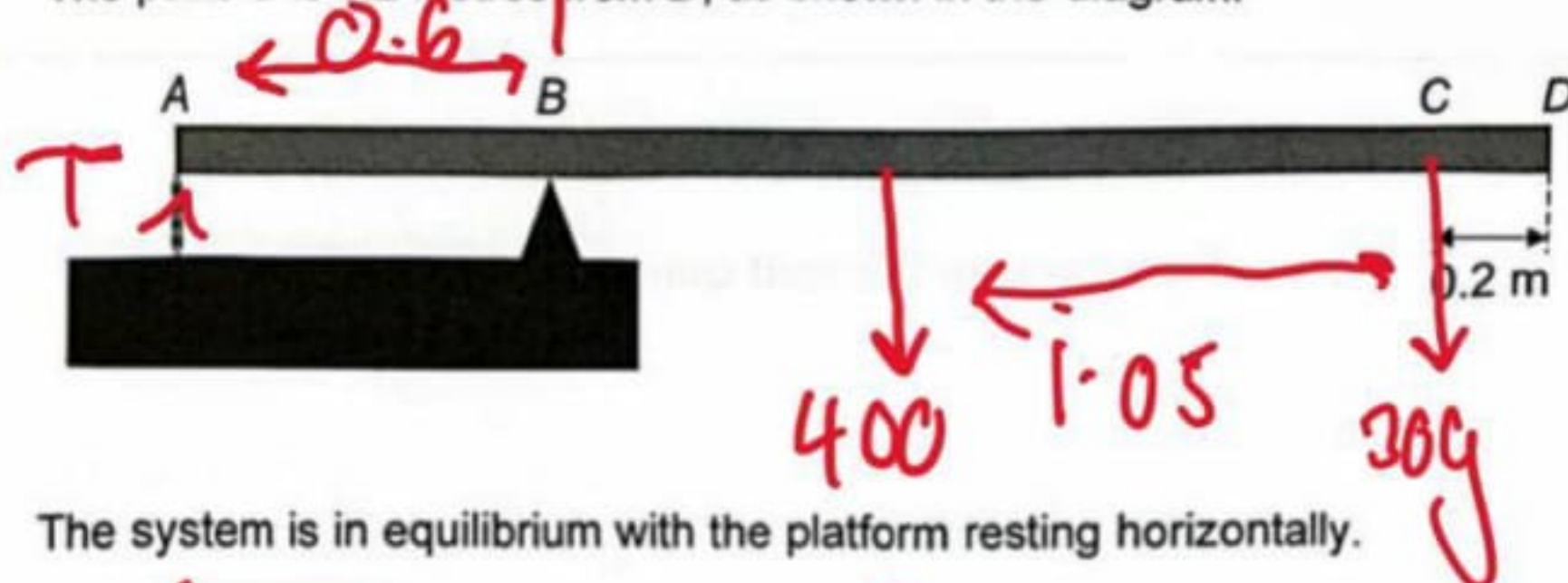
The platform is attached to a chain at A. The other end of the chain is fixed to the floor.

The platform rests on a support at a point B which is 0.6 metres from A

The support exerts an upwards reaction force on the platform.

A child of mass 30 kilograms stands at a point C

The point C is 0.2 metres from D, as shown in the diagram.



The system is in equilibrium with the platform resting horizontally.

18 (a)

By taking moments about A, find the reaction force at B

[4 marks]

$$(0.6 \times R) - (1.25 \times 400) - (2.3 \times 30g) = 0$$

$$0.6R - 500 - 69g = 0$$

$$R = \frac{500 + 69g}{0.6} = 1961.5$$

18 (b) (i) The child moves from C to B

State what happens to the reaction force at B during this movement.

[1 mark]

Reaction force would increase.

18 (b) (ii) Having reached B , the child steps off the platform.

State what happens to the tension in the chain during this movement.

[1 mark]

tension remains fixed.

Turn over for the next question

19

The displacement, s metres, of a particle P , at time t seconds, is given by

$$s = (2t^3)\mathbf{i} + (2t^2 + qt)\mathbf{j}$$

19 (a)

Find an expression for the velocity of particle P after t seconds.

[2 marks]

$$V = \frac{ds}{dt}$$

$$\frac{ds}{dt} = (6t^2)\mathbf{i} + (4t + q)\mathbf{j}$$

19 (b)

The acceleration, $a \text{ m s}^{-2}$, of a particle Q , at time t seconds, is given by

$$a = 6t\mathbf{i} + 7\mathbf{j}$$

Particle Q has an initial velocity of $4\mathbf{j} \text{ m s}^{-1}$ Particles P and Q are moving parallel to each other when $t = 2$ Find the value of q

$$V = \int a dt \quad \int (6t\mathbf{i} + 7\mathbf{j}) dt = (3t^2)\mathbf{i} + (7t)\mathbf{j} + C$$

[6 marks]

$$\text{When } t=0 \quad V=4\mathbf{j}, \quad 4\mathbf{j} = 0\mathbf{i} + 0\mathbf{j} + C$$

$$C = 4\mathbf{j}$$

$$V(\text{particle } Q) = (3t^2)\mathbf{i} + (7t)\mathbf{j} + 4\mathbf{j}$$

$$V(\text{particle } P) = (6t^2)\mathbf{i} + (4t + q)\mathbf{j}$$

$$\text{When } t=2, \text{ Velocity } Q: 12\mathbf{i} + 18\mathbf{j}, \text{ Velocity } P: 24\mathbf{i} + (8 + q)\mathbf{j}$$

$$\text{Therefore } 2Q = P, \quad 36 = 8 + q, \quad q = 28$$